

PUBLICATIONS BY EXPERIMENT

NOvA Collaboration

Refereed

- S. Abubakar et al. (NOvA Collaboration), “Measurement of π^0 Production in $\bar{\nu}_\mu$ Charged-Current Interactions in the NOvA Near Detector”, Phys. Rev. D 113, 112008 (2026)
- S. Abubakar et al. (NOvA Collaboration), “Ionization-based search for magnetic monopoles using the NOvA Far Detector”, Phys. Rev. D 113, 112014 (2026)
- M. A. Acero et al. (NOvA Collaboration), “Precision Measurement of Neutrino Oscillation Parameters with 10 Years of Data from the NOvA Experiment”, Phys. Rev. Lett. 136 (2026)
- M. A. Acero et al. (NOvA Collaboration), “Explanation of the seasonal variation of cosmic multiple muon events observed with the NOvA Near Detector”, Phys. Rev. D 113 (2026)
- M. A. Acero et al. (NOvA Collaboration), “Measurement of $d^2\sigma/d|q|dE_{avail}$ in charged current neutrino-nucleus interactions at $\langle E_\nu \rangle = 1.86$ GeV using the NOvA Near Detector”, Phys. Rev. D. 111, 052009 (2025)
- M. A. Acero et al. (NOvA Collaboration), “Dual-Baseline Search for Active-to-Sterile Neutrino Oscillations in NOvA”, Phys. Rev. Lett. 134, 081804 (2025)
- M. A. Acero et al. (NOvA and T2K Collaborations), “Joint neutrino oscillation analysis from the T2K and NOvA experiments”, Nature 646, 818 (2025)
- M. A. Acero et al. (NOvA Collaboration), “Monte Carlo method for constructing confidence intervals with unconstrained and constrained nuisance parameters in the NOvA experiment”, JINST 20 02, T02001 (2025)
- M. A. Acero et al. (NOvA Collaboration), “Search for CP-violating Neutrino Non-Standard Interactions with the NOvA Experiment”, Phys. Rev. Lett. 133 20, 201802 (2024)
- M. A. Acero et al. (NOvA Collaboration), “Expanding neutrino oscillation parameter measurements in NOvA using a Bayesian approach”, Phys. Rev. D 110, 012005 (2024)
- M. A. Acero et al. (NOvA Collaboration), “Measurement of the ν_e -Nucleus Charged-Current Double-Differential Cross Section at $\langle E_\nu \rangle = 2.4$ GeV using NOvA”, Phys. Rev. Lett. 130 (2023) 5, 051802 (2023)
- M. A. Acero et al. (NOvA Collaboration), “Improved measurement of neutrino oscillation parameters by the NOvA experiment”, Phys. Rev. D 106, 032004 (2022)
- M. A. Acero et al. (NOvA Collaboration), “Search for Active-Sterile Antineutrino Mixing Using Neutral-Current Interactions with the NOvA Experiment”, Phys. Rev. Lett. 127, 201801 (2021)
- M. A. Acero et al. (NOvA Collaboration), “First Measurement of Neutrino Oscillation Parameters using Neutrinos and Antineutrinos by NOvA”, Phys. Rev. Lett. 123 151803 (2019)
- M. A. Acero et al. (NOvA Collaboration), “New constraints on oscillation parameters from ν_e appearance and ν_μ disappearance in the NOvA experiment”, Phys. Rev. D 98, 032012 (2018)
- P. Adamson et al. (NOvA Collaboration), “Search for active-sterile neutrino mixing using neutral-current interactions in NOvA”, Phys. Rev. D. 96, 072006 (2017)
- P. Adamson et al. (NOvA Collaboration), “Constraints on Oscillation Parameters from ν_e Appearance and ν_μ Disappearance in NOvA”, Phys. Rev. Lett. 118, 231801 (2017)
- P. Adamson et al. (NOvA Collaboration), “Measurement of the neutrino mixing angle θ_{23} in NOvA”, Phys. Rev. Lett. 118, 151802 (2017)
- P. Adamson et al. (NOvA Collaboration), “First Measurement of Electron Neutrino Appearance in NOvA”, Phys. Rev. Lett. 116, 151806 (2016)
- P. Adamson et al. (NOvA Collaboration), “First Measurement of muon-neutrino disappearance in NOvA”, Phys. Rev. D 93, 051104(R) (2016)

Non-refereed

- S. Abubakar et al. (NOvA Collaboration), “The NOvA Test Beam Experiment”, arXiv:2609.10804 (2026)
- S. Abubakar et al. (NOvA Collaboration), “NOvA Dual-Baseline Search for Active-to-Sterile Neutrino Oscillations using Neutrino- and Antineutrino-Enriched Samples”, arXiv:2609.08900

(2026)

- S. Abubakar et al. (NOvA Collaboration), “Neutron detector response modeling in NOvA”, arXiv:2609.06956 (2026)
- S. Abubakar et al. (NOvA Collaboration), “Measurement of the $\bar{\nu}_\mu$ -Hydrogen Charged-Current Quasi-Elastic Cross Section using the NOvA Near Detector”, arXiv:2608.12293 (2026)
- S. Abubakar et al. (NOvA Collaboration), “Timing-Based Search for Magnetic Monopoles with the NOvA Detector on the Surface”, arXiv:2608.03888 (2026)
- S. Abubakar et al. (NOvA Collaboration), “Determining Neutrino Mass Ordering with NOvA and Upcoming JUNO Measurements”, arXiv:2606.14121 (2026)
- S. Abubakar et al. (NOvA Collaboration), “First inclusive triple-differential measurement of the muon-antineutrino charged-current cross section using the NOvA Near Detector”, arXiv:2603.06718 (2026)
- M. A. Acero et al. (NOvA Collaboration), “Measurement of the double-differential cross section of muon-neutrino charged-current interactions with low hadronic energy in the NOvA Near Detector”, arXiv:2410.10222 (2024)

DUNE Collaboration

Refereed

- S. Abbaslu et al. (DUNE Collaboration), “Reconstruction of atmospheric neutrinos in DUNE’s horizontal-drift far-detector module”, *Eur. Phys. J. C* 86 7, 904 (2026)
- S. Abbaslu et al. (DUNE Collaboration), “Charge readout electronics for the DUNE horizontal drift far detector: design and performance in ProtoDUNE-HD”, *JINST* 21 08, P08018 (2026)
- S. Abbaslu et al. (DUNE Collaboration), “Identification of low-energy kaons in the ProtoDUNE-SP detector”, *Phys. Rev. D* 113, 052004 (2026)
- S. Abbaslu et al. (DUNE Collaboration), “Operation of a Modular 3D-Pixelated Liquid Argon Time-Projection Chamber in a Neutrino Beam”, *Instruments* 10, 18 (2026)
- S. Abbaslu et al. (DUNE Collaboration), “Spatial and temporal evaluations of the liquid argon purity in ProtoDUNE-SP”, *JINST* 20 09, P09008 (2025)
- A. Abed Abud et al. (DUNE Collaboration), “The track-length extension fitting algorithm for energy measurement of interacting particles in liquid argon TPCs and its performance with ProtoDUNE-SP data”, *JINST* 20 02, P02021 (2025)
- A. Abed Abud et al. (DUNE Collaboration), “Supernova pointing capabilities of DUNE”, *Phys. Rev. D* 111 9, 092006 (2025)
- A. Abed Abud et al. (DUNE Collaboration), “First Measurement of the Total Inelastic Cross-Section of Positively-Charged Kaons on Argon at Energies Between 5.0 and 7.5 GeV”, *Phys. Rev. D* 110 9, 092011 (2024)
- A. Abed Abud et al. (DUNE Collaboration), “The DUNE Far Detector Vertical Drift Technology. Technical Design Report”, *JINST* 19 08, T08004 (2024)
- A. Abed Abud et al. (DUNE Collaboration), “DUNE Phase II: scientific opportunities, detector concepts, technological solutions”, *JINST* 19 12, P12005 (2024)
- A. Abed Abud et al. (DUNE Collaboration), “Performance of a Modular Ton-Scale Pixel-Readout Liquid Argon Time Projection Chamber”, *Instruments* 8 3, 41 (2024)
- A. Abed Abud et al. (DUNE Collaboration), “Doping liquid argon with xenon in ProtoDUNE Single-Phase: effects on scintillation light”, *JINST* 19 08, P08005 (2024)
- A. Abed Abud et al. (DUNE Collaboration), “Impact of cross-section uncertainties on supernova neutrino spectral parameter fitting in the Deep Underground Neutrino Experiment”, *Phys. Rev. D* 107, 112012 (2023)
- A. Abed Abud et al. (DUNE Collaboration), “Design, construction and operation of the ProtoDUNE-SP Liquid Argon TPC”, *JINST* 17 01, P01005 (2022)
- B. Abi et al. (DUNE Collaboration), “Supernova neutrino burst detection with the Deep Underground Neutrino Experiment”, *Eur. Phys. J. C* 81 5, 423 (2021)
- B. Abi et al. (DUNE Collaboration), “First results on ProtoDUNE-SP liquid argon time projection chamber performance from a beam test at the CERN Neutrino Platform”, *JINST* 15 12, P12004 (2020)
- B. Abi et al. (DUNE Collaboration), “Long-baseline neutrino oscillation physics potential of the DUNE experiment”, *Eur. Phys. J. C* 80 10, 978 (2020)
- B. Abi et al. (DUNE Collaboration), “Deep Underground Neutrino Experiment (DUNE), Far Detector Technical Design Report, Volume IV: Far Detector Single-phase Technology”, *JINST* 15 08, T08010 (2020)

Non-refereed

- S. Abbaslu et al. (DUNE Collaboration), “A Bayesian approach to the long-baseline neutrino oscillation sensitivity of DUNE”, *arXiv:2608.04059* (2026)
- S. Abbaslu et al. (DUNE Collaboration), “Operation and performance of ProtoDUNE Dual Phase liquid argon time projection chamber”, *arXiv:2607.15927* (2026)
- S. Abbaslu et al. (DUNE Collaboration), “Measurement of Exclusive π^+ -argon Interactions Using ProtoDUNE-SP”, *arXiv:2511.13462* (2025)
- S. Abbaslu et al. (DUNE Collaboration), “First Measurement of π^+ -Ar and p -Ar Total Inelastic Cross Sections in the Sub-GeV Energy Regime with ProtoDUNE-SP Data”, *arXiv:2511.11925*

(2025)

- A. Abed Abud et al. (DUNE Collaboration), “European Contributions to Fermilab Accelerator Upgrades and Facilities for the DUNE Experiment”, arXiv:2503.23744 (2025)
- A. Abed Abud et al. (DUNE Collaboration), “DUNE Software and Computing Research and Development”, arXiv:2503.23743 (2025)
- A. Abed Abud et al. (DUNE Collaboration), “The DUNE Science Program”, arXiv:2503.23291 (2025)
- A. Abed Abud et al. (DUNE Collaboration), “The DUNE Phase II Detectors”, arXiv:2503.23293 (2025)
- Abed Abud et al. (DUNE Collaboration), “DUNE Offline Computing Conceptual Design Report”, arXiv:2210.15665 (2022)
- A. Abed Abud et al. (DUNE Collaboration), “Snowmass Neutrino Frontier: DUNE Physics Summary”, Snowmass 2021, arXiv:2203.06100 (2022)
- B. Abi et al. (DUNE Collaboration), “Deep Underground Neutrino Experiment (DUNE), Far Detector Technical Design Report, Volume II: DUNE Physics”, arXiv:2002.03005 (2020)

T2K Collaboration

- D. Allan et al. (T2K Collaboration), “The Electromagnetic Calorimeter for the T2K Near Detector”, JINST 8 P10019 (2013)
- K. Abe et al. (T2K Collaboration), “First Muon-Neutrino Disappearance Study with an Off-Axis Beam”, Phys. Rev. D. 85, 031103 (2012)
- K. Abe et al. (T2K Collaboration), “Indication of Electron Neutrino Appearance from an Accelerator-produced Off-axis Muon Neutrino Beam”, Phys. Rev. Lett. 107, 041801 (2011)

Other

- T. K. Aarrestad et al. (HEP Community), “Building an AI-native Research Ecosystem for Experimental Particle Physics: A Community Vision”, arXiv:2602.17582 (2026)
- V. Daniel Elvira et al., “The Future of High Energy Physics Software and Computing”, Snowmass 2021, arXiv:2210.05822 (2022)
- G.S. Davies, P. Onyisi, and A. Roberts, “CompF5: End User Analysis Topical Group Report”, Snowmass 2021, arXiv:2209.14984 (2022)